Confidential Computing Network for cross-database DNA Relatives Matching

Pavel Nikonorov[1], Mariam Makhmutova[1], Andrew Ponomarev[1], Ruslan Vakhitov[1], Stanislav Nikolskiy[1] (1) GENXT, Wellcome Genome Campus, UK.

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https://genxt.network/

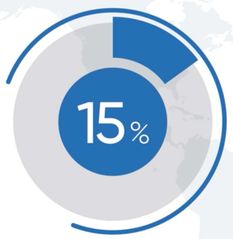
Background

DNA Relatives Matching and Ancestry prediction have been the key drivers in the personal genomics market, accounting for more than 50% of people who purchase direct-to-consumer genetic tests. There are at least 40 million DNA kits sold worldwide, demonstrating a significant potential for scientific advancements.

DNA RELATIVES MATCHING & ANCESTRY PREDICTION DRIVE PERSONAL GENOMICS

Problem

To identify potential relatives, Genetic Testing Companies (GTC) match their customers within their local databases. This approach, however, limits the value of this service. There are more than 250 GTCs worldwide and each may contain close and distant cousins.



OF GENETIC TESTING COMPANIES OFFER ANCESTRY & DNA MATCHING

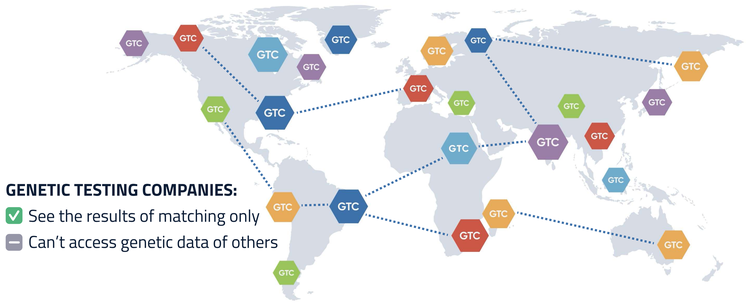
Moreover, most GTCs are unable to offer popular DNA services even locally due to limited database sizes. Our market research showed that only about 15% of the companies provide DNA Relatives Matching service.

GTCs are also limited to use local databases only due to privacy concerns arising in the context of potential data sharing scenarios and due to the absence of enterprise-grade federated infrastructure solutions ensuring both software integrity and data confidentiality.

Solution

GENXT has developed a peer-to-peer DNA Relatives Matching service for direct-to-consumer GTCs that allows them to cross-match their customers without any genomic data disclosure.

PEER-TO-PEER DNA RELATIVES MATCHING



GTCs exchange matched customers with the personal information that customers agreed to disclose before initiating the network search.

Confidential Computing

Confidential Computing protects data-in-use by performing computations in a hardware-based Trusted Execution Environment (TEE) which is essentially an isolated area of CPU memory.

Confidential Computing offers the ability to process data from multiple sources inside the TEE and get joint insights.

Provided by all major cloud providers, the technology guarantees that data is processed for intended purposes only and cannot be disclosed to any stakeholders both during and after computations.

THE LINUX FOUNDATION

CONFIDENTIAL COMPUTING CONSORTIUM

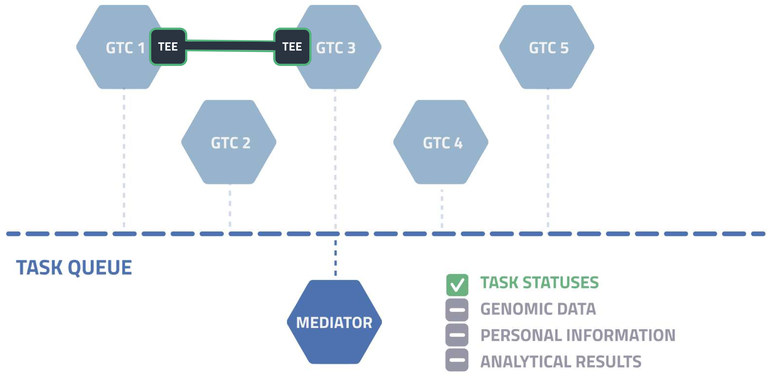
DNA Relatives Matching pipeline

GRAPE is a Genomic RelAtedness detection PipelinE that is reliable and accurate end-to-end software which is free for both commercial and academic purposes under GPLv3 license.

GRAPE combines data preprocessing, IBD segment recovery, masking of artifactual IBD segments, and accurate relationship estimation. In GRAPE's paper we demonstrate the pipeline's accuracy on a simulated population and provide a rationale for the reliability of our approach.

GRAPE was created to work with locally available genomic data banks, and was further adopted as a worker in the peer-to-peer confidential computing network.

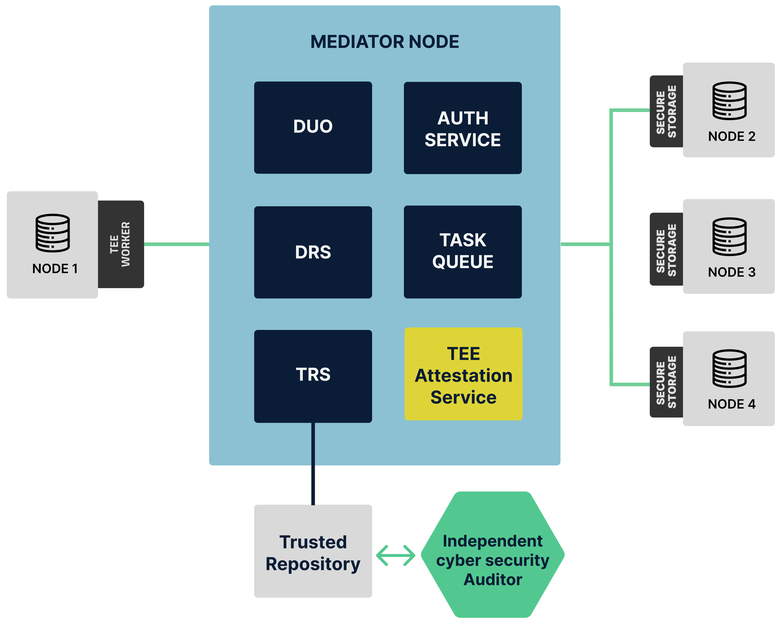
Mediated peer-to-peer network



Mediator Node is hosted by GENXT. It provides network discovery and hosts the Task Queue that contains task statuses only.

GA4GH standarts and frameworks

The solution employs Crypt4GH for storing and transferring data over end-to-end encryption channels between TEE-nodes whose genuineness and trustworthy is continuously attested, thereby keeping data always encrypted and available for consented processing only during the whole life-cycle.



The project is undergoing further development by adopting the following GA4GH standards: Service-Info Specification, Machine-Readable Consents, DUO, DRS, TRS, Phenopackets, WES, AAI & Passports to provide fine-grained access for research purposes.